

1 Solution — GIAM 8.3

Problem 4

1.1 Problem

The Barber’s paradox: *In a small town there is a Barber who shaves those men (and only those men) who do not shave themselves. Who shaves the Barber?* Explain the similarity to the proof of Cantor’s theorem.

1.2 The Barber’s Paradox Resolved

The town’s premise is *internally contradictory*. Examine the two possible answers:

- **Suppose the Barber shaves himself.** Then he belongs to the set of men he shaves. But the rule says he shaves only men who do *not* shave themselves — so he must *not* shave himself. That contradicts the supposition.
- **Suppose the Barber does not shave himself.** Then he belongs to the set of men who don’t shave themselves. But the rule says he shaves *every* such man — so he *does* shave himself. That also contradicts the supposition.

Both cases collapse. The resolution: *no such Barber can exist*. The premise — “there is a Barber who shaves exactly the men who don’t shave themselves” — describes a logically impossible town.

The Barber's Paradox: Either Answer Self-Destructs

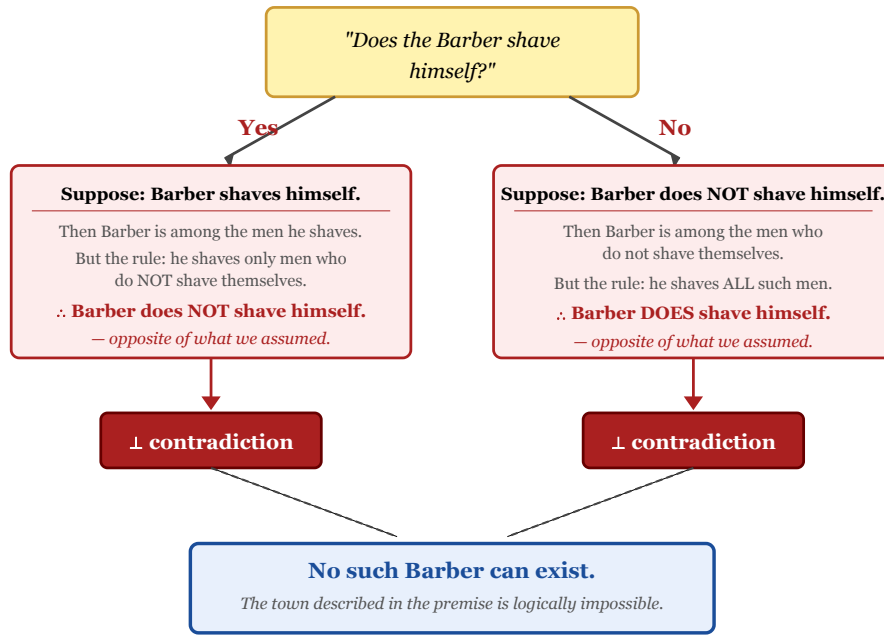


Figure 1.1: The Barber's paradox shown as a decision tree. Both answers to 'does Barber shave himself?' force the opposite of what was assumed, producing a contradiction in each branch. The only consistent conclusion is that no such Barber can exist.

1.3 The Cantor Argument Recapped

Cantor's theorem (Theorem 8.3.1) says: for any set A and any function $f : A \rightarrow \mathcal{P}(A)$, f cannot be surjective. The proof builds an explicit set $S \in \mathcal{P}(A)$ that is not in the image of f :

$$S = \{x \in A : x \notin f(x)\}.$$

Suppose for contradiction that $S \in \text{image}(f)$ — that is, $S = f(y)$ for some $y \in A$. Ask: *is* $y \in S$?

- If $y \in S$: by the definition of S , $y \notin f(y) = S$. Contradiction.
- If $y \notin S$: by the definition of S , $y \in f(y) = S$. Contradiction.

Either branch contradicts itself, so no such y exists; S is not in the image of f ; f is not surjective. ■

1.4 The Structural Parallel

The two arguments are not merely *similar* — they are the *same argument*, applied in two different settings. Both share the same five-step skeleton: (i) ambient universe, (ii) “flipping” diagonal set, (iii) hypothesis that something hits that set, (iv) self-referent question, (v) contradiction in both branches.

Barber’s Paradox ↔ Cantor’s Theorem	
<i>Both arguments share the same structure: self-reference applied to a “flipping” set.</i>	
Barber’s Paradox	Cantor’s Theorem
SETUP Town T ; Barber $b \in T$; b “shaves” some subset of T .	SETUP Set A ; element $y \in A$; $f : A \rightarrow \mathcal{P}(A)$ with $f(y) \subseteq A$.
DIAGONAL SET $\{ \text{men in } T \text{ who do NOT shave themselves} \}$	DIAGONAL SET $S = \{ x \in A : x \notin f(x) \}$
HYPOTHESIS Barber b shaves EXACTLY the men in the diagonal set.	HYPOTHESIS $f(y) = S$ for some $y \in A$, i.e., S lies in the image of f .
SELF-REFERENT QUESTION <i>“Does the Barber shave the Barber?”</i>	SELF-REFERENT QUESTION <i>“Is $y \in S$?” (i.e., is $y \in f(y)$?)</i>
OUTCOME (both branches) Yes $\Rightarrow$ rule excludes him. No $\Rightarrow$ rule includes him.	OUTCOME (both branches) $y \in S \Rightarrow y \notin f(y) = S$. $y \notin S \Rightarrow y \in f(y) = S$.
CONCLUSION No such Barber exists — the town is impossible.	CONCLUSION No such y exists — $S \notin \text{image}(f)$, so f isn’t surjective.

The Barber is exactly the “y” of Cantor’s proof — an entity whose existence the diagonal set rules out.

Figure 1.2: Side-by-side mapping of the Barber’s paradox to Cantor’s theorem. Each row shows a structural element in both arguments: setup, the ‘flipping’ diagonal set, the hypothesis that some entity hits that set, the self-referent question, the both-branches contradiction, and the conclusion that the hypothesized entity cannot exist.

In symbols, the dictionary between the two arguments is:

Barber’s Paradox	Cantor’s Theorem
Town T	Set A
Barber $b \in T$	Element $y \in A$
For each man $m \in T$, the set $\text{shaves}(m)$ of men m shaves	For each $x \in A$, the subset $f(x) \in \mathcal{P}(A)$
$b \in \text{shaves}(b)$?	$y \in f(y)$?

$D = \{m \in T : m \notin \text{shaves}(m)\}$ (“men who don’t shave themselves”)	$S = \{x \in A : x \notin f(x)\}$
Hypothesis: $\text{shaves}(b) = D$	Hypothesis: $f(y) = S$
Self-referent question: $b \in D?$	Self-referent question: $y \in S?$
Contradiction either way	Contradiction either way
Conclusion: no such b exists	Conclusion: no such y exists; f is not surjective

Table 1.1

Reading off the dictionary: the Barber’s paradox is *exactly* Cantor’s argument applied to the function $\text{shaves} : T \rightarrow \mathcal{P}(T)$ that sends each man m to the set $\text{shaves}(m)$ of men he shaves. The “Barber” is the element $y \in T$ that would witness $S \in \text{image}(\text{shaves})$; Cantor’s theorem says no such witness exists — i.e., the Barber cannot exist.

1.5 The Self-Reference Flip — A Common Recipe

What makes both arguments tick is a single trick: **construct a “diagonal” entity whose defining property *flips* when applied to itself.**

- Barber’s diagonal D : “set of men who don’t shave themselves.” The defining property of $m \in D$ is *negation* of $m \in \text{shaves}(m)$.
- Cantor’s diagonal S : “set of elements not in their own image under f .” The defining property of $x \in S$ is *negation* of $x \in f(x)$.

If a “diagonal-realizing” entity exists (b with $\text{shaves}(b) = D$, or y with $f(y) = S$), then asking the self-referent question — “does the entity have its own defining property?” — produces

$$\underbrace{b \in D}_{?} \iff \underbrace{b \notin \text{shaves}(b)}_{\text{def of } D} = \underbrace{b \notin D}_{\text{hypothesis}}, \quad \underbrace{y \in S}_{?} \iff \underbrace{y \notin f(y)}_{\text{def of } S} = \underbrace{y \notin S}_{\text{hypothesis}}.$$

In both cases, the statement and its negation become equivalent. This is a *logical impossibility* — no proposition P satisfies $P \Leftrightarrow \neg P$. The way out is to deny the hypothesis that the diagonal-realizing entity exists.

1.6 Remark — Russell’s Paradox Is the Same Argument

Russell’s paradox is the same construction applied to the “universal” function: in *naive* set theory, consider the would-be set

$$R = \{x : x \notin x\}.$$

Is $R \in R$?

- If $R \in R$: by the definition of R , $R \notin R$. Contradiction.
- If $R \notin R$: by the definition of R , $R \in R$. Contradiction.

The fix: R cannot be a set. This is Cantor’s argument applied to the would-be identity function $\text{id} : V \rightarrow \mathcal{P}(V)$ on the (would-be) class V of all sets, with $\text{id}(x) = x$. The diagonal subset $S = \{x : x \notin \text{id}(x)\} = \{x : x \notin x\} = R$; Cantor’s theorem says R is not in the image of id , but id ’s image is V itself, so $R \notin V$ — i.e., R is not a set. Modern set theory (ZFC) avoids the paradox by axiomatically restricting which collections count as sets.

So the three arguments — Barber, Cantor, Russell — share a common engine. The difference is what gets concluded:

Setting	Diagonal-realizing entity that can’t exist	What we conclude exists less than expected
Barber (town gossip)	a Barber with the stated rule	the town in the puzzle
Cantor (set theory)	a y mapped to S	a surjection $A \rightarrow \mathcal{P}(A)$
Russell (foundations)	the set $R = \{x : x \notin x\}$	a set of “all sets that don’t contain themselves”

Table 1.2

1.7 Remark — Why the Resolution Differs

The Barber’s paradox is *folk* mathematics — the conclusion (“no such Barber”) is uncontroversial and uninteresting, because nobody believes that arbitrary verbal descriptions of

towns must describe possible towns. The Barber’s paradox is a *puzzle* whose value is pedagogical: it makes the self-reference flip palpable.

Cantor’s theorem is *deep* mathematics — the conclusion (“ $\mathcal{P}(A)$ is strictly larger than A ”) is non-obvious and powerful. It gives the first proof that there are *different sizes* of infinity, and underlies the entire transfinite hierarchy $\aleph_0 < \aleph_1 < \aleph_2 < \dots$ explored later in the chapter.

Russell’s paradox is *foundational* — its discovery in 1901 forced a rebuild of set theory’s axioms (Zermelo–Fraenkel, 1908), restricting which collections count as sets so that the self-reference flip can no longer construct R . Without Russell’s paradox we wouldn’t have ZFC — the standard foundation of modern mathematics.

So the same diagonal trick is used at *three different levels of seriousness*: a puzzle (Barber), a theorem (Cantor), and an axiomatic revolution (Russell). The argumentative structure is identical in all three.

1.8 Remark — The Diagonal Trick Beyond Set Theory

The “construct a diagonal entity, ask if it has its own property, contradict either way” recipe shows up in many other fundamental results:

- *The halting problem* (Turing, 1936): there is no program H that, given any program p and input x , decides whether p halts on x . Proof: assume H exists; build $D(p)$ that runs $H(p, p)$ and halts iff H says p doesn’t halt on p . Ask: does $D(D)$ halt? Both branches contradict.
- *Gödel’s incompleteness theorems* (1931): in any sufficiently powerful consistent formal system, there is a statement G asserting “ G is not provable in the system.” Both G provable and $\neg G$ provable contradict consistency.
- *The Liar paradox*: the statement “This statement is false” cannot have a truth value — assigning either True or False makes it the other.
- *Tarski’s undefinability of truth* (1936): no formal language can define its own truth predicate.

In each case, a *self-referential* construction is set up so that any attempt to evaluate it flips the answer. The Barber’s paradox is the most intuitive entry point to this family of

arguments; Cantor's theorem is the first major mathematical result in the family; the others extend the trick into computation, formal logic, and the limits of definability.